Geometric Formulation and Thickness Modeling of Classical NACA Airfoils

(Independent Study Project)

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Abstract

Airfoil geometry is a fundamental determinant in aerodynamics, playing a crucial role in the performance, efficiency, and customized design of aircraft. The specific geometric characteristics of an airfoil directly influence its aerodynamic behavior, including lift, drag, and pitching moment coefficients. A clear understanding of basic airfoil design concepts is, therefore, essential for beginners in aerospace engineering. This article introduces the classical NACA airfoils and examines their geometric formulation using an Excel-based computational model.

Among the established airfoil families, the NACA 4-digit and 5-digit series, developed by the National Advisory Committee for Aeronautics (NACA), remain cornerstones in preliminary design, engineering education, and as benchmark profiles for advanced computational analyzes. This report aims to provide students interested in aerospace engineering with a structured understanding of the classical NACA airfoil families nomenclature and the governing geometric parameters, thereby offering both visual and technical insight into how the specified digits determine the resulting airfoil shape.

Introduction

Throughout the twentieth century, numerous airfoil families were introduced, including the Clark Y and the NACA 4-digit series. Over time, several thousand airfoils have been developed and experimentally tested. The Wright brothers themselves evaluated several hundred airfoils using a homemade wind tunnel, marking one of the earliest systematic investigations of wing profiles.

In the early years of aviation, aircraft wings relied heavily on external bracing systems, such as struts and wires, to provide structural support. As stronger construction materials became available, engineers sought to eliminate these external supports in order to reduce aerodynamic drag. Achieving this objective required thicker airfoil sections capable of providing sufficient structural strength, leading to the evolution of airfoil geometries similar to the Clark Y profile.

Early textbooks on flight theory frequently used the Clark Y airfoil to illustrate fundamental aerodynamic principles, and this practice has persisted over time. However, the airfoil families developed by the National Advisory Committee for Aeronautics (NACA) were documented more systematically. In particular, NACA Report No. 460 [3], published in November 1933 and entitled *The Characteristics of the Seventy-Eight Related Airfoil Sections from Tests in the Variable Density Wind Tunnel*, provided detailed experimental data for a series of related airfoils.

As a governmental research agency, NACA conducted extensive wind tunnel testing to determine the aerodynamic efficiency of various airfoil configurations under different flight conditions. NACA later evolved into the National Aeronautics and Space Administration (NASA), which continued research and development in airfoil design. The resulting NACA airfoil profiles were designated using a standardized numbering system consisting of four, five, or six digits, enabling a direct relationship between numerical parameters and geometric characteristics.

NACA Airfoil Theory

As early as the 1920s, research institutions in Europe and the United States began systematic measurements of the aerodynamic characteristics of airfoils in practical use. The results were organized into families of airfoils exhibiting specific aerodynamic properties. With a catalog of experimentally validated airfoils, aircraft designers were able to select suitable profiles efficiently for particular flight applications.

The National Advisory Committee for Aeronautics (NACA) conducted one of the most comprehensive and systematic investigations into the influence of airfoil ge-

ometry on aerodynamic performance. Early cambered airfoils, such as the Clark-Y and Göttingen sections, were known from initial wind tunnel experiments to exhibit favorable aerodynamic characteristics. NACA adopted these existing profiles as references; by removing the camber and normalizing the thickness-to-chord ratio, the resulting geometries became systematically comparable. A polynomial curve fit was then used to define the thickness distribution, forming the foundation of many subsequent NACA airfoil families, including the classic NACA 00-series symmetric airfoils.

The airfoil geometry is defined in a Cartesian coordinate system in which the x-coordinate is measured along the chord line and normalized by the chord length c. The leading edge corresponds to $x = 0$, while the trailing edge corresponds to $x = 1$.

The airfoil surfaces are generated by superimposing the thickness distribution $y_t(x)$ onto the mean camber line $y_c(x)$. The upper and lower surface coordinates are therefore defined as

$$\begin{aligned} x_u &= x - y_t \sin\theta, \\ y_u &= y_c + y_t \cos\theta, \end{aligned} \tag{1}$$

$$\begin{aligned} x_l &= x + y_t \sin\theta, \\ y_l &= y_c - y_t \cos\theta, \end{aligned} \tag{2}$$

where the local inclination angle θ of the mean camber line is given by

$$\theta = \tan^{-1}\left(\frac{dy_c}{dx}\right). \tag{3}$$

Classical Thickness Distributions

Matching Empirical Curves by Simultaneous Power Equations

Many important developments in aerodynamics have historically been empirical in nature. As theoretical frameworks advanced, it became necessary to express these empirical observations in analytical form. Notable examples include the thickness equation of the classical NACA four-digit symmetric airfoils, thin-airfoil theory formulations, and spanwise loading distributions.

In addressing such problems, mathematicians rely primarily on two powerful analytical tools: power series expansions and Fourier series representations. These methods allow experimentally observed geometric or aerodynamic trends to be approximated using continuous mathematical expressions, as mentioned by Alan Pope in *Basic Wing and Airfoil Theory* [7].

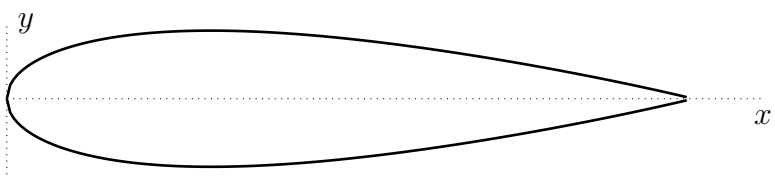

Figure 1: NACA 0020 profile

To illustrate this procedure, consider the development of the NACA 0020 airfoil equation. During the period from 1925 to 1935, two airfoils with markedly different camber characteristics were widely used: the Göttingen 398 and the Clark Y. When the camber was removed from these profiles, their thickness distributions were found to be nearly identical. This observation motivated the adoption of a common thickness distribution combined with various parabolic camber lines in order to systematically investigate the influence of maximum camber magnitude and its chordwise location.

A critical step in this process was the formulation of a mathematical expression for the basic symmetric thickness shape. The empirical geometric data corresponded to a 20 percent thick airfoil defined in a Cartesian coordinate system, with the leading edge located at the origin and the trailing edge at $x = 1$.

The empirical conditions were:

(1) Maximum ordinate

$$x = 0.3, \qquad y = 1.0, \qquad \frac{dy}{dx} = 0$$

(2) Ordinate at the trailing edge

$$x = 1.0, \qquad y = 0.002$$

(3) Trailing edge angle

$$x = 1.0, \qquad \frac{dy}{dx} = -0.234$$

(4) Nose shape

$$x = 0.1, \qquad y = 0.078$$

Inspection of the nose region suggests that the leading-edge shape may be approximated by a square-root function of the form $y = a_0\sqrt{x}$. However, to ensure closure at the trailing edge, a linear correction term may be introduced as a first approximation. This leads to the preliminary form

$$y = a_0\sqrt{x} + a_1 x.$$

To satisfy all five empirical conditions, higher-order polynomial terms are added, yielding the general expression

$$y = a_0\sqrt{x} + a_1 x + a_2 x^2 + a_3 x^3 + a_4 x^4.$$

Differentiation gives

$$\frac{dy}{dx} = \frac{1}{2}a_0 x^{-1/2} + a_1 + 2a_2 x + 3a_3 x^2 + 4a_4 x^3.$$

With five empirical constraints and five unknown coefficients, the resulting system of simultaneous equations can be solved to obtain

$$y = 0.2969\sqrt{x} - 0.1260x - 0.3516x^2 + 0.2843x^3 - 0.1015x^4.$$

For airfoils of other thickness ratios within the same family, the ordinates of the 0.20 thickness-ratio model are scaled by the factor $(t/0.20)$, where t is the ratio of the maximum thickness to the chord, c.

$$\pm\, y_t = \left(\frac{t}{0.2}\right)\left(0.2969\sqrt{x} - 0.1260x - 0.3516x^2 + 0.2843x^3 - 0.1015x^4\right)$$

The leading-edge radius of this airfoil family is defined as the radius of curvature of the basic thickness equation evaluated at $x = 0$. Owing to the presence of the term $a_0\sqrt{x}$, the radius of curvature remains finite at the leading edge and can be shown to be

$$R_{LE} = \frac{a_0^2}{2}\left(\frac{t}{0.2}\right)^2 \equiv 1.1019\, t^2, \tag{4}$$

which is obtained by taking the limit as $x \to 0$ of the standard expression for the radius of curvature,

$$R = \frac{\left[1 + \left(\dfrac{dy}{dx}\right)^2\right]^{3/2}}{\left|\dfrac{d^2y}{dx^2}\right|}.$$

The trailing-edge angle is defined as the angle between the upper and lower surfaces at the trailing edge. It may be expressed as

$$\delta_{TE} = 2\tan^{-1}\left[\frac{t}{0.2}\left(\frac{a_0}{2} + a_1 + 2a_2 + 3a_3 + 4a_4\right)\right] \equiv 2\tan^{-1}(1.0625\, t), \tag{5}$$

which is obtained by evaluating the slope of the thickness distribution at $x = 1$ (or equivalently $x = c$ in dimensional form).

The local surface inclination at the trailing edge is therefore

$$\varphi = \left| \frac{dy_t}{dx} \right| \Bigg|_{x=1} .$$

Because the upper and lower surfaces are symmetric with respect to the mean camber line for a symmetric airfoil, the total trailing-edge angle is twice this value, i.e., $\delta_{TE} = 2\varphi$.

Matching Empirical Curves by Simultaneous Fourier Series

A second approach for representing empirical aerodynamic curves employs trigonometric series consisting of sine and cosine terms. In many cases, this method is more efficient and mathematically convenient than using exponential relations.

Fourier series are particularly useful in aerodynamic theory because they allow complex distributions to be approximated by a finite number of harmonic terms. Two classical applications illustrate this approach. The first application involves representing a camber line that is not symmetric about the mid-chord. In this case, a sine series is typically employed since sine terms naturally satisfy boundary conditions at the leading and trailing edges while allowing asymmetric curvature. The second application concerns the representation of lift distribution along a finite wing that is symmetric about the mid-span. In this case, a cosine series is commonly used, as cosine functions inherently capture symmetric distributions about the central axis.

By using simultaneous Fourier series, empirical aerodynamic curves can be matched with high accuracy while preserving essential physical boundary conditions. This approach forms an important theoretical foundation in thin airfoil theory and lifting-line theory.

Reformulation of Thickness Distribution for Sharp Trailing Edge

The classical NACA 4-digit thickness distribution is defined using a polynomial representation with a small but finite trailing edge thickness. For the original formulation, the trailing edge ordinate at $x = 1$ is slightly different from zero due to the coefficient of the x^4 term (-0.1015), resulting in a finite trailing edge.

In the present work, the geometric condition at the trailing edge is modified to enforce a sharp trailing edge:

$$x = 1, \qquad y = 0.$$

To satisfy this boundary condition, the coefficient of the x^4 term is recalculated such that

$$0.2969 - 0.1260 - 0.3516 + 0.2843 - a_4 = 0.$$

Solving for a_4 gives:

$$a_4 = 0.1036.$$

Thus, the thickness distribution for a symmetric airfoil with maximum thickness ratio $t = 0.20$ becomes:

$$y(x) = 0.2969\sqrt{x} - 0.1260x - 0.3516x^2 + 0.2843x^3 - 0.1036x^4.$$

For airfoils with other thickness ratios within the same family, the ordinate distribution is scaled linearly with respect to the maximum thickness ratio:

$$\pm y_t(x) = \left(\frac{t}{0.2}\right)\left(0.2969\sqrt{x} - 0.1260x - 0.3516x^2 + 0.2843x^3 - 0.1036x^4\right).$$

This modification ensures a geometrically sharp trailing edge while preserving the overall thickness distribution characteristics of the original NACA formulation.

NACA 4-Digit Series

The NACA 4-digit series represents the first family of airfoils developed through a systematic mathematical approach to correlate geometry with aerodynamic performance. The designation system defines the airfoil through three primary parameters: the first digit specifies the maximum camber as a percentage of the chord; the second digit indicates the chordwise location of this maximum camber in tenths of the chord from the leading edge; and the final two digits represent the maximum thickness as a percentage of the chord, as described in NACA Report No. 460 [3].

Thus, the NACA 2412—a quintessential wing section for general aviation— features a maximum camber of 2 percent of the chord located at 0.4 of the chord from the leading edge, with a maximum thickness of 12 percent of the chord. Symmetric airfoils constitute a critical subset of this family where the mean camber line coincides exactly with the chord line. Because the maximum camber is zero, the first two digits are denoted as "00," rendering the camber position irrelevant. The NACA 0012 is the industry standard for symmetric sections, widely utilized for horizontal and vertical stabilizers due to its zero pitching moment at a zero angle of attack and its predictable aerodynamic behavior. For simulations requiring a thinner profile to model high-speed tail surfaces, the NACA 0009 serves as a primary representative.

Mean Camber Line Formulation

The mean camber line of the four-digit series is defined by two parabolic segments of the general form

$$y = b_0 + b_1 x + b_2 x^2.$$

The constants are determined from the following boundary conditions:

(1) Camber-line endpoints

$$\begin{aligned} x &= 0, & y &= 0 \\ x &= 1, & y &= 0 \end{aligned}$$

(2) Maximum camber condition

$$x = p, \qquad y = m$$

Thus, the mean camber line is given by

$$y_c = \begin{cases} \dfrac{m}{p^2}(2px - x^2), & 0 \le x \le p \\ \dfrac{m}{(1-p)^2}\left[(1-2p) + 2px - x^2\right], & p \le x \le 1 \end{cases} \tag{6}$$

The slope of the camber line is obtained by differentiation:

$$\frac{dy_c}{dx} = \begin{cases} \dfrac{2m}{p^2}(p - x), & 0 \le x \le p \\ \dfrac{2m}{(1-p)^2}(p - x), & p \le x \le 1 \end{cases} \tag{7}$$

In practical applications, the parameters m, p, and t may be varied to tailor the aerodynamic characteristics of the airfoil. Increasing the maximum camber generally enhances the lift coefficient, while the camber location influences the pitching moment and pressure distribution. The thickness ratio affects structural strength, drag characteristics, and leading-edge radius.

By systematically varying these parameters, a wide range of airfoils can be generated within the four-digit series to meet specific design requirements.

NACA 5-Digit Series

Following the success of the four-digit family, the NACA 5-digit series was developed to push the boundaries of maximum lift capability and aerodynamic efficiency. While the series retains the thickness distribution of its predecessor, it introduces a significantly more complex mean camber line. The primary design objective was to shift the maximum camber substantially forward, as experimental data indicated that forward camber placement could drastically increase the maximum lift coefficient ($C_{l_{max}}$) while simultaneously reducing the pitching moment (C_m) and minimum profile drag. The geometry and aerodynamic characteristics of the five-digit family are studied in NACA Report No. 537 [6].

The five-digit designation system reflects these performance-oriented goals. The first digit represents the design lift coefficient (C_{l_i}) in tenths, multiplied by 3/2. The second and third digits together define the position of the maximum camber, and the final two digits indicate the maximum thickness ratio. For simulations, the 230-series is the quintessential representative of a "high-lift, low-moment" section.

A specialized sub-family within this series incorporates reflexed camber lines, where the trailing edge curves slightly upward. These sections were mathematically derived to produce a near-zero pitching moment about the quarter-chord, making them theoretically ideal for tailless or flying-wing configurations where longitudinal stability is a primary concern. The NACA 23112 serves as a primary representative of this reflexed geometry. While providing unique stability benefits, these sections have seen more limited practical application compared to the standard 230-series due to a slight reduction in maximum lift potential.

Three-Digit Mean Camber Line

To obtain a camber line with a strongly forward maximum camber location, the so-called three-digit camber formulation was introduced in NACA Report No. 537 [6].

The camber line is defined in two regions such that the curvature (second derivative) decreases linearly to zero at a specified junction point $x = m$ and remains zero from that point to the trailing edge.

Thus,

For $0 \leq x \leq m$: $\dfrac{d^2y}{dx^2} = k_1(x - m)$

For $m \leq x \leq 1$: $\dfrac{d^2y}{dx^2} = 0$

The design constraints are:

(1) Camber-line endpoints

$$\begin{aligned} x &= 0, & y &= 0 \\ x &= 1, & y &= 0 \end{aligned}$$

(2) Continuity at the junction point $x = m$

$$y_N = y_T$$
$$\left(\frac{dy}{dx}\right)_N = \left(\frac{dy}{dx}\right)_T$$

where the subscripts $_N$ and $_T$ refer to the fore and aft equations, respectively.

The resulting camber line is

$$y_c = \begin{cases} \dfrac{k_1}{6}\left[x^3 - 3mx^2 + m^2(3-m)x\right], & 0 \le x \le m \\ \dfrac{k_1 m^3}{6}(1-x), & m \le x \le 1 \end{cases} \tag{8}$$

and the slope of the simple camber is

$$\frac{dy_c}{dx} = \begin{cases} \dfrac{k_1}{6}\left[3x^2 - 6mx + m^2(3-m)\right], & 0 \le x \le m \\ -\dfrac{k_1 m^3}{6}, & m \le x \le 1. \end{cases} \tag{9}$$

The parameter m is determined such that the maximum camber occurs at chordwise positions p calculated by multiplying the second digit by 0.05. Consequently, digits 1 through 5 correspond to standard maximum camber locations at $0.05c$, $0.10c$, $0.15c$, $0.20c$, and $0.25c$, respectively.

The location of maximum camber is obtained from the condition

$$\left.\frac{dy_c}{dx}\right|_{x=p} = 0,$$

which yields

$$p = m\left[1 - \sqrt{\frac{m}{3}}\right].$$

The constant k_1 is selected to ensure the design lift coefficient (C_{l_i}) equals the product of the first digit and 0.15. Accordingly, first digits of 1, 2, and 3 correspond to ideal lift coefficients of 0.15, 0.30, and 0.45, respectively.

The tabulated parameters are as follows.

Camber-line designation	p	m	k_1
210	0.05	0.0580	361.400
220	0.10	0.1260	51.640
230	0.15	0.2025	15.957
240	0.20	0.2900	6.643
250	0.25	0.3910	3.230

Table 1: Tabulated parameters of non-reflexed camber lines for $C_{l_i} = 0.3$

Reflexed Mean Camber Line

To obtain a zero pitching moment about the quarter-chord, a reflexed mean line was introduced. In this formulation, the curvature is again defined piecewise:

For $0 \leq x \leq m$: $\dfrac{d^2y}{dx^2} = k_1(x - m)$

For $m \leq x \leq 1$: $\dfrac{d^2y}{dx^2} = k_2(x - m)$

Solving the related equations yields

$$y_c = \begin{cases} \dfrac{k_1}{6}\left[(x-r)^3 - \dfrac{k_2}{k_1}(1-r)^3 x - r^3 x + r^3\right], & 0 \leq x \leq r \\ \dfrac{k_1}{6}\left[\dfrac{k_2}{k_1}(x-r)^3 - \dfrac{k_2}{k_1}(1-r)^3 x - r^3 c + r^3\right], & r \leq x \leq 1 \end{cases} \tag{10}$$

The slope is derived using differentiation

$$\frac{dy_c}{dx} = \begin{cases} \dfrac{k_1}{6}\left[3(x-r)^2 - \dfrac{k_2}{k_1}(1-r)^3 - r^3\right], & 0 \leq x \leq r \\ \dfrac{k_1}{6}\left[3\dfrac{k_2}{k_1}(x-r)^2 - \dfrac{k_2}{k_1}(1-r)^3 - r^3\right], & r \leq x \leq 1 \end{cases} \tag{11}$$

Imposing the condition of a zero slope at the maximum camber location,

$$\left.\frac{dy_c}{dx}\right|_{x=p} = 0,$$

leads to

$$\frac{k_2}{k_1} = \frac{3(m-p)^2 - m^3}{(1-m)^3}.$$

For each prescribed value of p, the parameter m was determined to satisfy the condition of zero quarter-chord pitching moment ($C_{m_{c/4}} = 0$). Subsequently, k_1 was evaluated to produce a design lift coefficient of $C_{l_i} = 0.3$.

The tabulated values of m, k_1, and k_2/k_1 are summarized above.

Camber-line designation	p	m	k_1	k_2/k_1
211	0.05	-	-	-
221	0.10	0.1300	51.990	0.000764
231	0.15	0.2170	15.793	0.00677
241	0.20	0.3180	6.520	0.0303
251	0.25	0.4410	3.191	0.1355

Table 2: Tabulated coefficients of reflexed camber for $C_{l_i} = 0.3$

In practical design applications, the defining parameters of the five-digit series are not restricted to the tabulated integer values. Instead, they may be treated as continuous variables to achieve specific aerodynamic requirements. By adjusting the camber magnitude, camber location, and thickness ratio, designers can tailor the lift characteristics, pitching moment behavior, and drag performance to meet particular mission profiles. Thus, although originally defined by discrete designations, the five-digit formulation provides a flexible analytical framework for systematic airfoil development.

NACA 4- and 5-Digit Modified Series

The modified NACA four- and five-digit airfoil families were developed to provide greater geometric flexibility while preserving the basic thickness characteristics of the original series.

In the modified designation, the airfoil number is followed by a dash and a two-digit suffix (e.g., NACA 0012-63). The first digit after the dash represents the leading edge radius index, and the second digit indicates the location of maximum thickness in tenths of the chord measured from the leading edge, which is represented in NACA Report No. 492 [8].

As in the original four-digit series, the ordinates vary linearly with the thickness–chord ratio. Thus, for any desired thickness ratio t/c, the ordinates are obtained by scaling the design thickness distribution proportionally. The original NACA four-digit series is a special case of the modified formulation corresponding to: leading edge index $I = 6$ (normal leading edge) and maximum thickness location at $0.30c$. Hence, the classical four-digit airfoil can be interpreted as a particular instance of the more general modified family.

Thickness Distributions

The thickness distribution is defined piecewise so that the leading-edge radius and the chordwise location of maximum thickness may be varied independently.

Similar to the thickness distributions formulation of NACA 4-digit and 5-digit series, a critical step in this process was the formulation of a mathematical expression for the basic symmetric thickness shape. The empirical geometric data corresponded to a 20% thick airfoil defined in a Cartesian coordinate system.

If the chord is taken at the position x axis from 0 to 1, the ordinates y from the leading edge to the maximum thickness location ($0 \leq x \leq m$):

$$\pm\, y_t = a_0\sqrt{x} + a_1 x + a_2 x^2 + a_3 x^3,$$

From the maximum thickness location to the trailing edge ($m \leq x \leq 1$):

$$\pm\, y_t = d_0 + d_1(1-x) + d_2(1-x)^2 + d_3(1-x)^3.$$

The constants are determined from the following geometric constraints:

(1) Specified maximum thickness

$$x = m \qquad y = 0.10 \qquad \frac{dy}{dx} = 0$$

(2) Prescribed leading-edge radius

$$x = 0 \qquad R = \frac{a_0^2}{2}$$

(3) Specified radius of curvature at the point of maximum thickness

$$x = m \qquad R = \frac{(1-m)^2}{2d_1(1-m) - 0.588}$$

(4) Specified trailing-edge ordinate

$$x = 1 \qquad y = d_0 = 0.002$$

(5) Prescribed trailing-edge angle

$$x = 1 \qquad \frac{dy}{dx} = d_1 = f(m)$$

These conditions ensure continuity of ordinate, slope, and curvature at the junction point $x = m$.

m	d_1
0.2	0.200
0.3	0.234
0.4	0.315
0.5	0.465
0.6	0.700

Table 3: Selected values of d_1 to prevent curvature reversal [8]

Some approximations of the coefficients above were described by William H.Mason in his book *Lecture Notes on Configuration Aerodynamics* [4] as follows.

The Riegels approximate d_1 is define by

$$d_1 = f(m) \cong \frac{(2.24 - 5.42m + 12.3m^2)}{10(1 - 0.878m)}$$

Once the value of d_1 is known, d_2 and d_3 are found from the relations given by Riegels:

$$d_2 = \frac{0.294 - 2(1-m)d_1}{(1-m)^2}$$

and

$$d_3 = \frac{-0.196 + (1-m)d_1}{(1-m)^3}$$

With the d_i coefficients determined, the a_i coefficients can be found. First, a_0 is based on the leading edge radius:

$$\begin{aligned} a_0 &= 0.2969\,\frac{I}{6}, \quad \text{for } I \leq 8 \\ &= 0.2969\sqrt{3}, \quad \text{for } I = 9 \end{aligned}$$

Define the following

$$\rho_1 = \frac{1}{5}\frac{(1-m)^2}{[0.588 - 2d_1(1-m)]}$$

then the remaining coefficients can be found from:

$$a_1 = \frac{0.3}{m} - \frac{15}{8}\cdot\frac{a_0}{m} - \frac{m}{10\rho_1}$$

$$a_2 = -\frac{0.3}{m^2} + \frac{5}{4}\cdot\frac{a_0}{m^{3/2}} + \frac{1}{5\rho_1}$$

$$a_3 = \frac{0.1}{m^3} - \frac{0.375a_0}{m^{5/2}} - \frac{1}{10\rho_1 m}$$

Reformulated Thickness Distribution (Sharp Trailing Edge)

For the classical finite trailing-edge formulation,

$$x = 1 \qquad y = d_0 = 0.002.$$

To enforce a geometrically sharp trailing edge, the boundary condition is modified to

$$x = 1 \qquad y = d_0 = 0.$$

The coefficients d_2 and d_3 are obtained from the geometric constraints at $x = m$ and $x = 1$. For the general case,

$$d_2 = \frac{3(0.1 - d_0) - 2(1-m)d_1}{(1-m)^2}$$

$$d_3 = \frac{2(d_0 - 0.1) + (1-m)d_1}{(1-m)^3}$$

Substituting $d_0 = 0$ for the sharp trailing edge gives

$$d_2 = \frac{0.3 - 2(1-m)d_1}{(1-m)^2}$$

$$d_3 = \frac{-0.2 + (1-m)d_1}{(1-m)^3}$$

This modification affects only the aft thickness distribution. Since the leading-edge geometry and the maximum thickness conditions remain unchanged, the coefficients a_i derived previously are preserved.

Leading-Edge Radius and Trailing-Edge Angle

The leading-edge index I controls the magnitude of the leading-edge radius and is proportional to the coefficient a_0.

The relationship between the leading-edge radius R_{LE} and the index number I is given by

$$R_{LE} = \frac{1}{2}\left(0.2969\,\frac{t}{0.2}\,\frac{I}{6}\right)^2 .$$

Type	Leading-edge Index I	a_0
Sharp	0	0
Quarter-normal	3	0.148450
Normal	6	0.296900
Three-times normal	9	0.514246

Table 4: Leading-edge index classification

The trailing-edge angle depends on the position of maximum thickness, m, and is given by

$$\delta_{TE} = 2\tan^{-1}(d_1) = 2\tan^{-1}[f(m)]$$

Modified Five-Digit Series

The modified five-digit airfoils retain the same camber-line definitions as the standard five-digit series but incorporate the modified thickness formulation described above.

Thus, the camber line is defined according to the five-digit design equations, while the thickness distribution is replaced by the modified polynomial expressions that allow independent control of: leading-edge radius, maximum thickness location, and trailing-edge angle. This combination provides increased flexibility in tailoring airfoil geometry for specific aerodynamic performance requirements.

In practical aerodynamic design, the modified formulation allows the defining parameters to be treated as continuous variables rather than discrete tabulated values. By adjusting the camber magnitude, camber location, thickness ratio, leading-edge radius, and maximum thickness position, designers can systematically optimize lift characteristics, drag performance, and pitching moment behavior. Consequently, the modified NACA families provide a versatile analytical framework for preliminary airfoil design.

NACA 1-Series Airfoils and 16-Series Designation

Unlike previous airfoil families defined by geometric polynomials, the NACA 1-Series was developed using inverse design methods rooted in advanced aerodynamic theory. By the late 1930s, research shifted from prescribing geometry to specifying a desired pressure distribution and mathematically deriving the corresponding shape. This approach allowed designers to tailor the airfoil to specific performance targets, such as lift-to-drag ratios and stall behavior, rather than relying on empirical geometric iterations. Consequently, 1-Series airfoils are generated numerically rather than through simple analytical expressions, as documented in NACA Report No. 824 [5].

The identification of the 1-Series follows a five-digit convention, exemplified by the NACA 16-212. The initial digit "1" denotes the series, while the second digit indicates the chordwise location of minimum pressure in tenths of the chord; in this instance, "6" signifies that minimum pressure occurs at $0.6c$. Following the hyphen, the third digit represents the design lift coefficient in tenths ($C_{L,design} = 0.2$), and the final two digits indicate the maximum thickness ratio ($t/c = 0.12$).

The 16-Series constitutes the most practically significant subset of this family, optimized to maintain favorable pressure gradients over an extended portion of the chord. By shifting the minimum pressure point further aft, these airfoils reduce the suction peak at the leading edge, thereby delaying drag divergence and boundary layer separation at high subsonic speeds. While the thickness distribution of the 16-Series shares similarities with the modified 4-digit series, specifically regarding leading-edge radii and mid-chord maximum thickness, its camber line is strictly dictated by aerodynamic requirements [8]. Due to their widespread adoption in high-speed applications, the 16-Series is often categorized independently in contemporary aeronautical literature.

NACA 6-Series and 6A-Series

While earlier NACA iterations utilized approximate theoretical methods, they often failed to precisely achieve the pressure distributions required for significant drag reduction. The development of the NACA 6-Series represented a transition toward rigorous inverse airfoil theory. By prescribing an ideal pressure distribution first and then deriving the corresponding geometry, engineers sought to maximize the extent of the laminar boundary layer. This design philosophy focuses on maintaining a favorable pressure gradient over a substantial portion of the chord, thereby minimizing skin-friction drag and producing the characteristic "low-drag bucket", a discrete range of lift coefficients (C_l) where the airfoil operates with exceptional efficiency.

The 6-Series designation reflects this complexity, as seen in the example NACA 64_1-212, $a = 0.6$. The first digit "6" identifies the series, while the second digit ("4")

locates the minimum pressure at $0.4c$. The subscript ("1") indicates the width of the low-drag bucket, specifically maintaining laminar flow over a range of ± 0.1 about the design lift coefficient. Following the hyphen, the digit "2" specifies a design lift coefficient (C_{l_i}) of 0.2, and the final two digits ("12") denote a thickness ratio (t/c) of 0.12. The parameter a defines the fraction of the chord over which the pressure gradient remains favorable; for $a = 0.6$, this gradient is maintained over 60 percent of the chord [5].

NACA 6A-Series

The 6A-Series was subsequently developed to address the manufacturing and structural limitations of the original 6-Series. In the standard 6-Series, the theoretical derivation often resulted in a thin, concave "cusp" near the trailing edge, which was difficult to fabricate and prone to structural failure. The 6A-series modifies this by eliminating the trailing edge cusp, resulting in a geometry that is nearly straight from approximately 80 percent chord to the trailing edge. This modification simplifies production and improves the pitching moment characteristics while preserving most of the laminar flow benefits.

Camber Line

Unlike the four-digit and five-digit airfoil families, the camber line of the NACA 6-Series airfoils is not defined using simple polynomial expressions. Instead, the camber line is determined using inverse aerodynamic design methods to achieve the specified pressure distribution corresponding to the desired design lift coefficient. The camber distribution is calculated numerically using thin airfoil theory and potential flow analysis.

The mean lines commonly used with the NACA 6-series airfoils produce a uniform chordwise loading from the leading edge to the point $x = a$ and a linearly decreasing load from this point to the trailing edge. Data for NACA mean lines with values of a equal to 0, 0.1, 0.2, 0.3, 0.4, 0.5, 0.6, 0.7, 0.8, 0.9, and 1.0 are presented in the supplementary figures. The ordinates were computed using the following formula, which represents a simplification of the original expression for mean-line ordinates given NACA Report No. 824 [5]:

$$
\begin{aligned}
y = \frac{C_{l_i}}{2\pi(1+a)} \Bigg\{ & \frac{1}{1-a} \Bigg[\frac{1}{2}(a-x)^2 \ln(a-x) \\
& - \frac{1}{2}(1-x)^2 \ln(1-x) + \frac{1}{4}(1-x)^2 - \frac{1}{4}(a-x)^2 \Bigg] \\
& - x \ln x + g - hx \Bigg\}
\end{aligned}
$$

where

$$
g = -\frac{1}{1-a} \left[a^2 \left(\frac{1}{2} \ln a + \frac{1}{4} \right) + \frac{1}{4} \right]
$$

$$
h = \frac{1}{1-a} \left[\frac{1}{2}(1-a)^2 \ln(1-a) - \frac{1}{4}(1-a)^2 \right] + g
$$

and

$$
\frac{dy}{dx} = \frac{C_{l_i}}{2\pi(1+a)} \left\{ \frac{1}{1-a} \left[(1-x) \ln(1-x) - (a-x) \ln(a-x) \right] - \ln x - 1 - h \right\}
$$

The associated angle of attack is

$$
\alpha_i = \frac{C_{l_i} h}{2\pi(1+a)}
$$

Thickness Distributions

Similarly, the thickness distribution of the NACA 6-series airfoils was derived using theoretical aerodynamic methods rather than empirical geometric fitting. The design procedure was based on potential flow theory and conformal mapping techniques from complex analysis to generate airfoil shapes capable of producing prescribed pressure distributions. The derivation methods are introduced in NACA Report No. 824 [5], and a detailed theoretical analysis is presented in *Basic Wing and Airfoil Theory* [7] and *Theory of Wing Sections* [1].

The mathematical foundation of this design approach relies on conformal transformation techniques, which enable the mapping of simple analytical flow solutions into airfoil geometries with specific aerodynamic characteristics. These transformations preserve the governing equations of inviscid, irrotational flow while reshaping the geometry. As a result, they allow designers to control pressure gradients, ensure smooth pressure recovery, and promote extended laminar flow over a significant portion of the airfoil surface.

One of the classical derivation methods in thin airfoil theory is the Joukowski transformation, a conformal mapping used to generate the Joukowski airfoil from a circular cylinder in the complex plane. Although the NACA 6-series airfoils are not simple Joukowski profiles, the theoretical framework is conceptually similar. Conformal mapping techniques provide the analytical basis for determining thickness distributions that correspond to specified pressure distributions.

Methodology

Excel Graph (2D)

Microsoft Excel is one of the simplest tools for generating graphs and calculating tabulated data. This approach is particularly useful for understanding preliminary concepts in aircraft design, especially wing profiles, which are fundamental components of an airplane.

Using cosine spacing with 1000 points provides an efficient method for distributing coordinates along the chord. The primary advantage of cosine spacing is its ability to increase the density of points in regions with high curvature, thereby producing a smoother and more accurate airfoil representation.

There are two commonly used cosine spacing methods:

(1) The x-coordinates are defined using a sequence of angles ranging from 0 to $\frac{\pi}{2}$. This method concentrates more points near the leading edge, where the curvature is greatest, while fewer points are distributed near the trailing edge.

(2) The x-coordinates are defined using angles ranging from 0 to π. This method distributes points symmetrically, ensuring equal point density near both the leading edge and trailing edge. However, the points near the mid-chord region are more sparsely distributed compared to the edge regions.

Since 1000 points are sufficient to produce a smooth graphical representation, both cosine spacing methods yield nearly identical results. In this study, the first method is adopted.

Airfoil Graphing

For geometry-based thickness distributions, including the NACA 4-digit, 5-digit, and their modified series, the airfoil geometries are constructed using the analytical

equations provided in the official NACA reports. However, for airfoils designed based on thin airfoil theory, no explicit analytical thickness distributions are defined. Due to the complexity of aerodynamic analysis and the requirement for advanced computational tools, these airfoils are generated using available airfoil coordinate data. The governing equations are then approximated using Lagrange interpolation.

The geometric characteristics of airfoils vary significantly depending on changes in key parameters. These geometric variations directly influence aerodynamic performance, particularly lift, drag, and moment coefficients during flight.

In this study, one geometric parameter of a given NACA airfoil is varied while all other parameters are held constant. This approach allows for a clear observation of how individual parameters affect the airfoil shape.

Numerical Accuracy Assessment

Scope of the Error Analysis

The present tool does not involve CFD or flow-field simulation. The geometry is generated from analytical NACA formulations and, in some cases, is reconstructed from tabulated coordinate data using polynomial interpolation. Therefore, the primary sources of numerical error arise from:

1. Discretization error due to finite chordwise sampling.
2. Interpolation error from polynomial reconstruction.
3. Tabulated data approximation and rounding.

Since the analytical NACA equations are available, they are treated as the reference solution for validation purposes.

Discretization Error

Let $y_{exact}(x)$ denote the analytical airfoil surface and $y_N(x)$ the discretized representation using N chordwise points.

The discretization error is evaluated using:

$$E_\infty(N) = \max_i |y_N(x_i) - y_{exact}(x_i)|$$

and the root-mean-square (RMS) error:

$$E_{RMS}(N) = \sqrt{\frac{1}{N}\sum_{i=1}^{N}\left(y_N(x_i) - y_{exact}(x_i)\right)^2}$$

A grid refinement study is performed by increasing $N = 50, 100, 200$.

For smooth analytical airfoil functions, the discretization error is expected to decrease approximately with $\mathcal{O}(N^{-2})$ for sufficiently smooth regions.

Effect of Node Distribution

Uniform spacing and cosine spacing are compared under identical N.

Since airfoil curvature is highest near the leading edge, uniform spacing under-resolves this region, leading to locally amplified errors.

Cosine spacing clusters nodes near $x = 0$, thus significantly reducing leading-edge geometric error. For typical NACA 4-digit profiles, the maximum geometric deviation near the leading edge is expected to decrease by one order of magnitude when cosine spacing is used instead of uniform spacing at moderate resolution ($N \approx 100$).

Interpolation Error

When reconstructing geometry from tabulated data, Lagrange polynomial interpolation of degree n is applied.

The interpolation error is theoretically given by:

$$f(x) - P_n(x) = \frac{f^{(n+1)}(\xi)}{(n+1)!}\prod_{i=0}^{n}(x - x_i)$$

for some ξ in the interpolation interval.

Since airfoil thickness and camber distributions are smooth functions, the interpolation error remains small for moderate n. However, high-degree interpolation under uniform spacing may produce oscillations (Runge phenomenon).

In practical implementation, interpolation is restricted to a moderate polynomial degree or applied locally to avoid instability.

The expected interpolation error magnitude, for $n \leq 10$ and smooth tabulated data, is typically below $10^{-4}c$ in geometric coordinates.

Tabulated Data Approximation

Tabulated airfoil data are usually rounded to 4–5 decimal places. This introduces a baseline numerical uncertainty on the order of $E_{table} \sim 10^{-5}c$ to $10^{-4}c$, which represents the lower bound of achievable accuracy when reconstructing geometry from such data.

Overall Expected Accuracy

Combining discretization and interpolation effects, the total geometric error may be estimated as:

$$E_{total} \approx E_{disc} + E_{interp} + E_{table}$$

For typical configurations with $N \geq 100$ cosine-spaced nodes and a moderate interpolation degree, the overall geometric deviation is expected to remain within $E_{total} \leq 10^{-4}c$, corresponding to less than 0.01% of the chord length.

Acceptable Error and Adjustment Criteria

Since the present tool is intended for geometric visualization, preliminary aerodynamic estimation, and educational or conceptual analysis, a geometric tolerance of $E_{tol} = 10^{-4}c$ is considered acceptable.

If the error exceeds this threshold, favorable adjustments are: increase chordwise resolution, use cosine spacing instead of uniform spacing, reduce the interpolation degree or switch to spline interpolation, refine tabulated input precision.

Conclusion

The dominant numerical error source in the present implementation is discretization near the leading edge. Cosine spacing effectively mitigates this issue.

Under typical settings, the geometric error remains sufficiently small for engineering-level preliminary airfoil analysis. Further refinement is only necessary if high-fidelity CFD or structural analysis is to be performed using the generated geometry.

Influence of Design Parameters on Aircraft Wing Performance

The evolution of airfoil design, as documented in seminal works such as NACA TR-610 [2], demonstrates a systematic approach to correlating geometric definitions with aerodynamic performance. By examining the fundamental NACA families, ranging from the early 4-digit series to the laminar-flow 6-series, designers can tailor aircraft performance through specific parametric adjustments.

The NACA 4-Digit Series: Geometric Fundamentals

The NACA 4-digit series utilizes three geometric parameters: maximum camber, camber position, and maximum thickness. Increasing the maximum camber enhances the curvature of the mean camber line, which directly raises the lift coefficient at a zero angle of attack and increases the magnitude of the negative pitching moment about the quarter-chord. While thin airfoil theory suggests that camber primarily shifts the lift curve upward without altering its slope, excessive camber can induce adverse pressure gradients, leading to premature boundary layer separation.

Representative Profiles

NACA 0012 (Symmetrical standard for stabilizers), NACA 2412 (Commonly used in general aviation wings like the Cessna 172), and NACA 4415 (High-lift variant for heavy utility aircraft).

The NACA 5-Digit Series: Optimized Loading

The NACA 5-digit series represents a shift toward theoretical aerodynamic loading. Unlike the 4-digit series, the camber line is derived to meet a specific design lift coefficient, with the maximum camber typically positioned further forward. This forward shift significantly enhances maximum lift capabilities while maintaining lower pitching moments compared to 4-digit counterparts. A unique feature of this series is the reflex indication (the third digit), where a reflexed camber line, characterized by upward curvature at the trailing edge, is employed to neutralize nose-down pitching tendencies, a critical requirement for tailless or flying-wing configurations.

Representative Profiles

NACA 23012 (Renowned for its high lift-to-moment ratio, used on the B-17 and Beechcraft King Air) and NACA 23112 (Reflexed version for specialized stability).

Modified 4- and 5-Digit Series: High-Speed Refinement

The modified series introduces additional control over the leading-edge radius and the chordwise position of maximum thickness. Adjusting the leading-edge radius allows designers to balance stall characteristics against high-speed efficiency; a larger radius reduces sensitivity to angle-of-attack variations and delays separation, whereas a sharper radius may optimize specific drag regimes. Moving the position of maximum thickness, traditionally at 30 percent of the chord, further aft (to 40 percent or 50 percent) is a primary strategy for delaying drag divergence and improving high-subsonic performance, though it often necessitates a trade-off in structural spar depth.

Representative Profiles

NACA 0009-64 (Used in high-speed jet tail surfaces) and NACA 23012-64 (Modified for improved Mach number performance in early jet designs like the F-84).

NACA 1-Series and 16-Series: High-Speed Specialization

While the 1-series was the precursor to laminar flow research, the 16-series became the benchmark for high-speed, low-pressure applications. These airfoils are designed to maintain a nearly uniform pressure distribution over the chord, effectively eliminating the high-suction peaks that lead to flow separation or cavitation. Their geometry is characterized by a very small leading-edge radius and maximum thickness located near 50 percent of the chord, making them ideal for high-speed propellers and hydrofoils.

Representative Profiles

NACA 16-012 (Standard for high-speed propeller sections) and NACA 16-212 (Cambered version for racing aircraft).

NACA 6 and 6A Series: Laminar Flow Innovation

The NACA 6-series airfoils are engineered to maximize the extent of the laminar boundary layer by maintaining a favorable pressure gradient over a large portion of the chord. This results in the ”drag bucket” phenomenon, a specific range of lift coefficients where profile drag is exceptionally low. The design lift coefficient dictates the center of this bucket, while thickness influences both the width of the bucket and the structural integrity of the wing. The 6A series represents a practical modification where the trailing-edge ”cusp” is removed to simplify manufacturing and improve pitching moment behavior without significant loss of laminar efficiency.

Representative Profiles

NACA 64-212 (Classic laminar flow wing) and NACA 64A210 (Modernized version for high-performance jets like the F-16).

Results and Conclusion

Excel Model Capabilities

The developed Excel-based simulation tool provides a parametric visualization of NACA airfoil geometries. The model allows real-time generation of the airfoil profile based on user-defined input parameters within a designated control panel.

The tool is capable of:

- Displaying the complete airfoil geometry (the upper and lower surfaces).
- Plotting the mean camber line for cambered airfoils.
- Indicating the leading-edge radius based on the selected series formulation.
- Updating the airfoil profile dynamically as geometric parameters are modified.

The geometry responds directly to variations in defining parameters such as maximum camber, camber position, and thickness ratio. Any change in the control area automatically updates the plotted airfoil, enabling rapid comparative analysis between different configurations.

This parametric structure allows efficient investigation of geometric trends and facilitates understanding of how individual parameters influence airfoil shape. Although

the tool focuses primarily on geometric modeling rather than full aerodynamic performance prediction, it provides a clear and practical platform for preliminary airfoil design and educational analysis.

Comparison and Assessment

This study summarizes the evolution of NACA airfoils, from early geometry-based definitions to modern aerodynamic and empirical design approaches. Initially, airfoil design relied primarily on geometric parameters such as camber, camber position, and thickness, which allowed engineers to generate airfoil shapes using analytical equations. These early methods provided a foundation for understanding airfoil geometry and its influence on aerodynamic performance.

However, with the advancement of aerodynamic theory and computational tools, airfoil design has gradually shifted toward performance-based approaches. Modern airfoil development focuses on achieving specific aerodynamic objectives, such as target lift coefficients, favorable pressure distributions, and improved lift-to-drag ratios.

Simulation also plays a critical role in airfoil analysis, although numerical results may contain errors due to discretization, interpolation, numerical approximation, and modeling assumptions. Therefore, simulation results must be carefully validated and interpreted.

In practical applications, the digits in the NACA series are not strictly followed during manufacturing. Although airfoils may retain their NACA designation, their geometries are often refined through aerodynamic analysis, numerical simulations, and experimental validation. These refinements ensure that the final airfoil meets performance, stability, and efficiency requirements for specific applications such as aircraft wings, propellers, or turbine blades.

Discussion

Classical Design Approach

The classical airfoil design approach was primarily theoretical and geometry-driven. Engineers defined airfoil shapes using analytical equations without detailed aerodynamic performance predictions. The aerodynamic characteristics were then evaluated through experiments and empirical observations, followed by iterative modifications.

Although this approach established important theoretical foundations and provided valuable design concepts, it often resulted in inefficient designs and a relatively high failure ratio. The lack of accurate predictive tools made it difficult to optimize airfoils for specific aerodynamic requirements.

Modern Design Approach

Modern airfoil design relies heavily on numerical methods and computational tools. Airfoils are now designed to meet specific aerodynamic performance targets, such as desired lift coefficients C_l, lift-to-drag ratios, and pressure or velocity distributions along the chord.

The modern design process typically follows these steps:

1. Define aerodynamic performance requirements
2. Generate airfoil geometry using numerical optimization methods
3. Simulate aerodynamic performance using computational tools
4. Manufacture prototypes
5. Conduct experimental validation
6. Refine and optimize the design based on results

This approach significantly improves design reliability, efficiency, and performance compared to classical methods.

Modern airfoils are no longer defined solely by geometric parameters but by aerodynamic performance objectives. The development of computational fluid dynamics (CFD) and numerical optimization allows engineers to design airfoils with precise aerodynamic characteristics.

Limitations of 2D Airfoil Analysis and Transition to 3D Design

Airfoil sections represent only a two-dimensional approximation of wing behavior. The aerodynamic characteristics of a 2D airfoil correspond to those of an infinite wing with an infinite aspect ratio, which does not exist in practical applications.

In reality, wings, propeller blades, and turbine blades are three-dimensional and finite structures. Their aerodynamic behavior is influenced by additional factors

such as wingtip vortices, induced drag, and spanwise flow. As a result, accurate aerodynamic analysis requires three-dimensional numerical simulations.

Fortunately, modern aerospace engineering provides advanced computational tools such as OpenFOAM, OpenVSP, and ANSYS, which enable engineers to perform detailed aerodynamic simulations and optimize designs efficiently.

The development of such tools reflects the interdisciplinary nature of modern aerospace engineering. Their creation requires expertise in aerodynamics, applied mathematics, numerical methods, and computer science. This interdisciplinary collaboration has significantly improved the accuracy, efficiency, and reliability of airfoil and aircraft design.

Appendix

Cosine Spacing and Half-Cosine Spacing

Motivation

In airfoil geometry discretization, a uniform spacing in the chordwise direction often fails to accurately capture regions with high curvature, especially near the leading edge. Since the leading edge radius is small and the curvature is large, clustering points in this region improves geometric resolution and numerical stability.

To achieve non-uniform but smoothly distributed points, cosine-based spacing techniques are commonly used.

Cosine Spacing

Let the chord length be normalized such that $x \in [0, 1]$. The cosine spacing transformation is defined as:

$$x(\theta) = \frac{1}{2}(1 - \cos\theta) \quad \text{for } \theta \in [0, \pi]$$

If N points are required, we define:

$$\theta_i = \frac{i\pi}{N-1}, \quad i = 0, 1, \ldots, N-1$$

Thus,

$$x_i = \frac{1}{2}\left(1 - \cos\frac{i\pi}{N-1}\right)$$

Properties

- Point clustering occurs near both $x = 0$ (leading edge) and $x = 1$ (trailing edge).
- The distribution is symmetric with respect to $x = 0.5$.
- The spacing becomes denser near the endpoints.

Half-Cosine Spacing

In some applications, clustering is only required near the leading edge. In this case, the half-cosine transformation is used:

$$x(\theta) = 1 - \cos\theta \quad \text{for } \theta \in \left[0, \frac{\pi}{2}\right]$$

For N points:

$$\theta_i = \frac{i}{N-1} \cdot \frac{\pi}{2}, \quad i = 0, 1, \ldots, N-1$$

Thus,

$$x_i = 1 - \cos\left(\frac{i}{N-1} \cdot \frac{\pi}{2}\right)$$

Properties

- Strong clustering near $x = 0$.
- Nearly uniform spacing toward $x = 1$.
- Particularly suitable for airfoil leading edge refinement.

Comparison with Uniform Spacing

For reference, uniform spacing is defined as $x_i = \dfrac{i}{N-1}$

Uniform spacing does not account for geometric curvature and may require a significantly larger number of points to achieve comparable leading-edge accuracy. Cosine-based spacing provides higher geometric resolution with the same number of discretization points.

Lagrange Interpolation

Definition

Given a set of $n+1$ distinct data points:

$$(x_0, y_0), (x_1, y_1), \ldots, (x_n, y_n)$$

the Lagrange interpolation polynomial $P_n(x)$ of degree at most n is defined as:

$$P_n(x) = \sum_{j=0}^{n} y_j L_j(x)$$

where $L_j(x)$ are the Lagrange basis polynomials:

$$L_j(x) = \prod_{\substack{m=0 \\ m \neq j}}^{n} \frac{x - x_m}{x_j - x_m}$$

Interpolation Property

The basis polynomials satisfy:

$$L_j(x_i) = \begin{cases} 1, & i = j \\ 0, & i \neq j \end{cases}$$

Thus,

$$P_n(x_i) = y_i$$

which guarantees exact interpolation at the given nodes.

Application in Airfoil Geometry

In airfoil reconstruction, Lagrange interpolation can be used to:

- Interpolate discrete airfoil coordinate data.
- Construct smooth curves from control points.
- Approximate camber line or thickness distribution functions.

However, for large n, high-degree interpolation may suffer from Runge's phenomenon, especially under uniform spacing. Using cosine-spaced nodes significantly improves numerical stability.

Error Estimation

The interpolation error is given by:

$$f(x) - P_n(x) = \frac{f^{(n+1)}(\xi)}{(n+1)!} \prod_{i=0}^{n} (x - x_i)$$

for some ξ in the interval containing the nodes.

This expression shows that:

- The error depends on node placement.
- Clustering nodes (e.g., cosine spacing) reduces oscillatory behavior.

Representatives of Classical NACA Airfoils

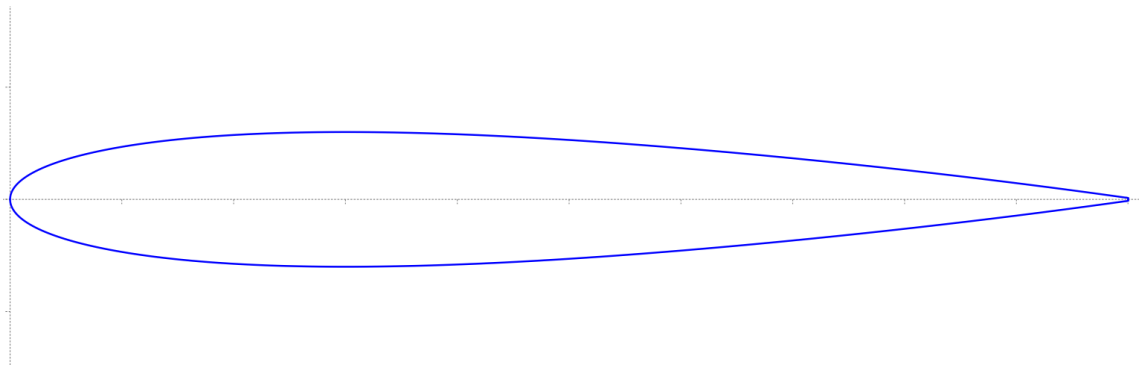

Figure 2: NACA 0012

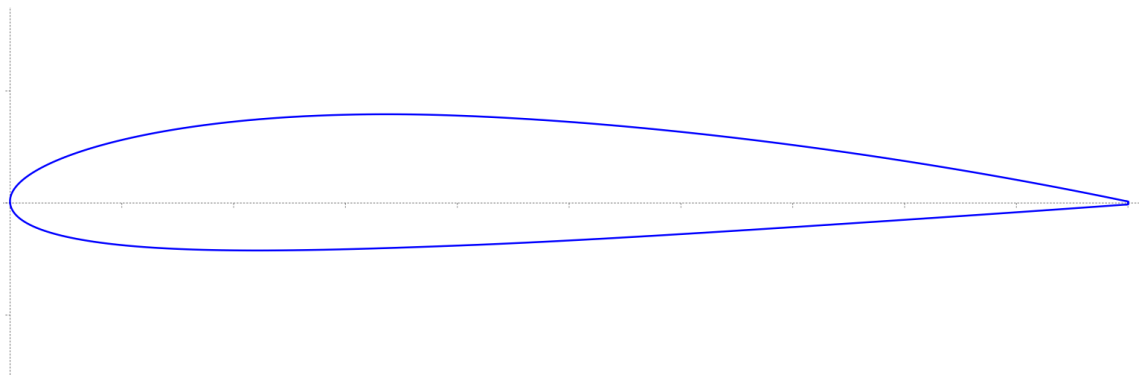

Figure 3: NACA 2412

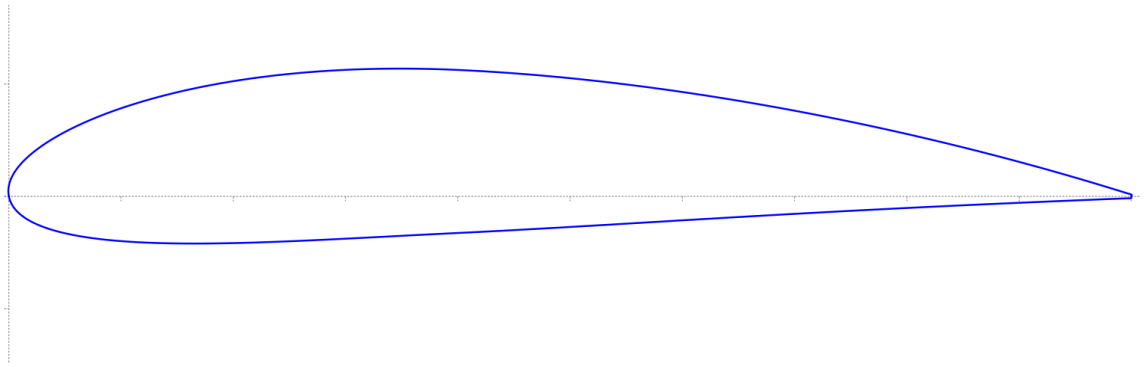

Figure 4: NACA 4415

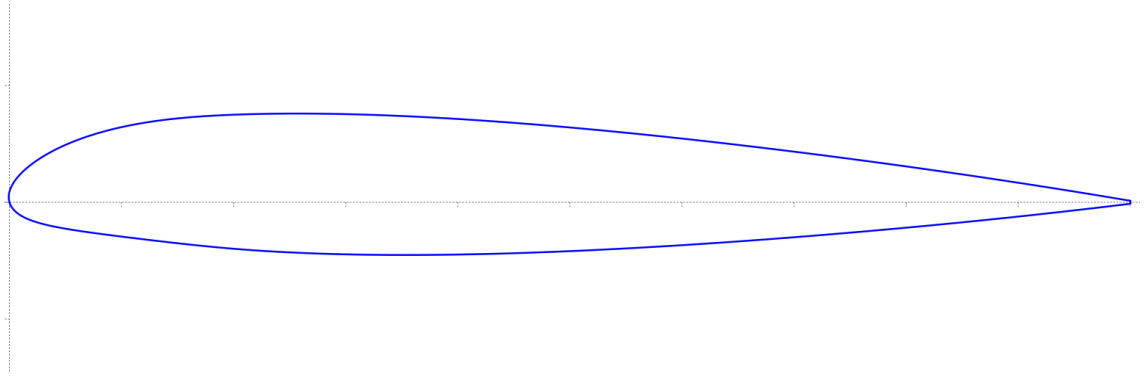

Figure 5: NACA 23012

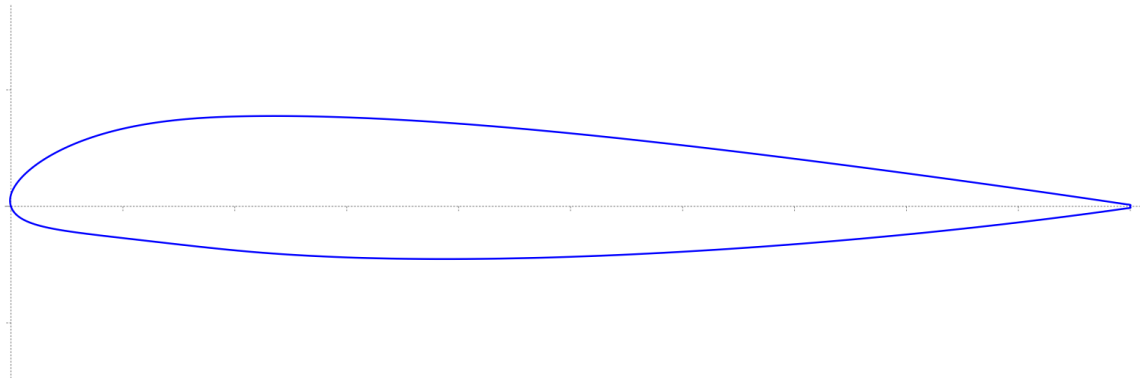

Figure 6: NACA 23112

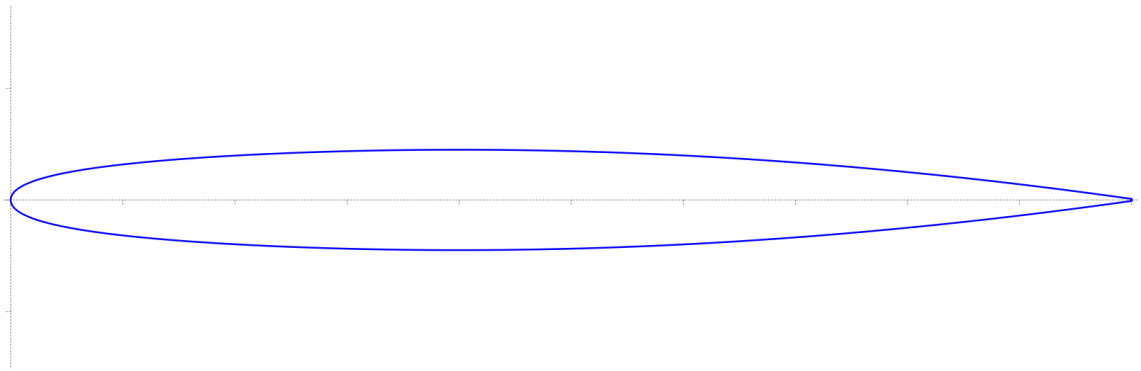

Figure 7: NACA 0009-64

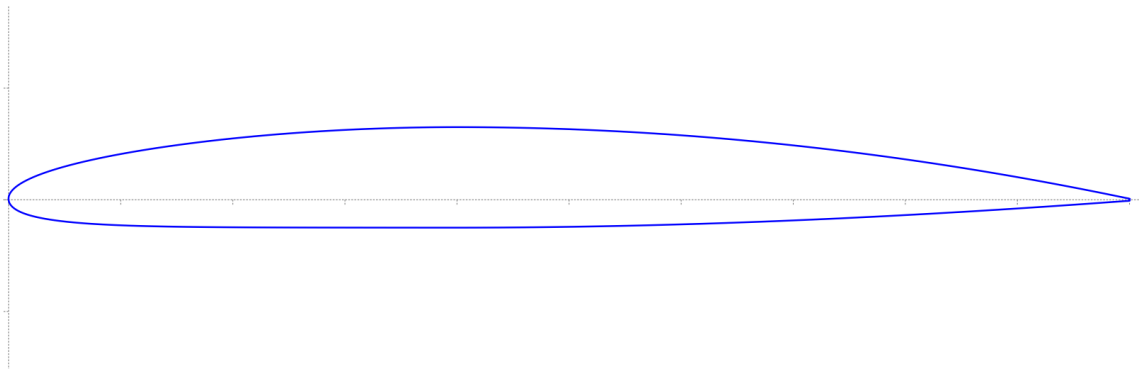

Figure 8: NACA 2409-64

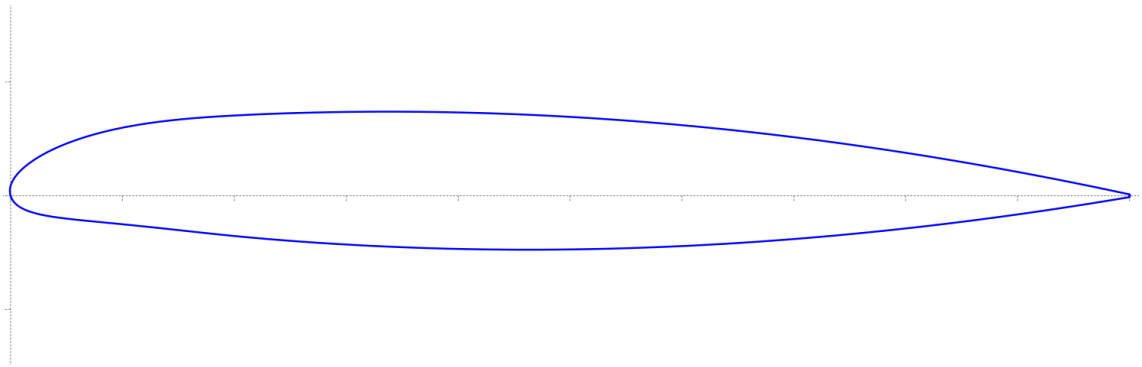

Figure 9: NACA 23012-64

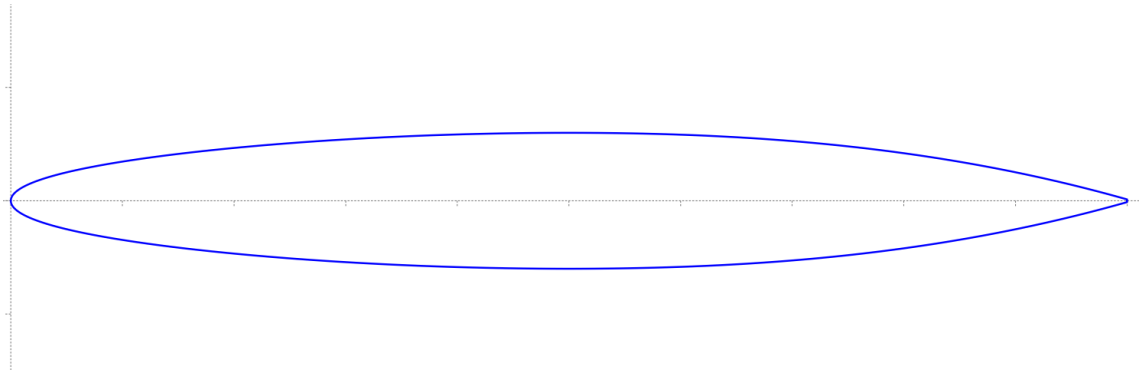

Figure 10: NACA 16-012

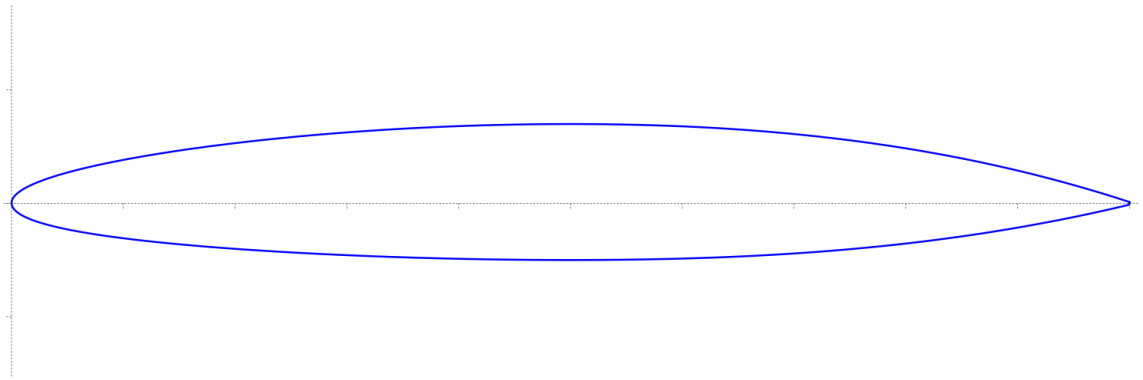

Figure 11: NACA 16-212

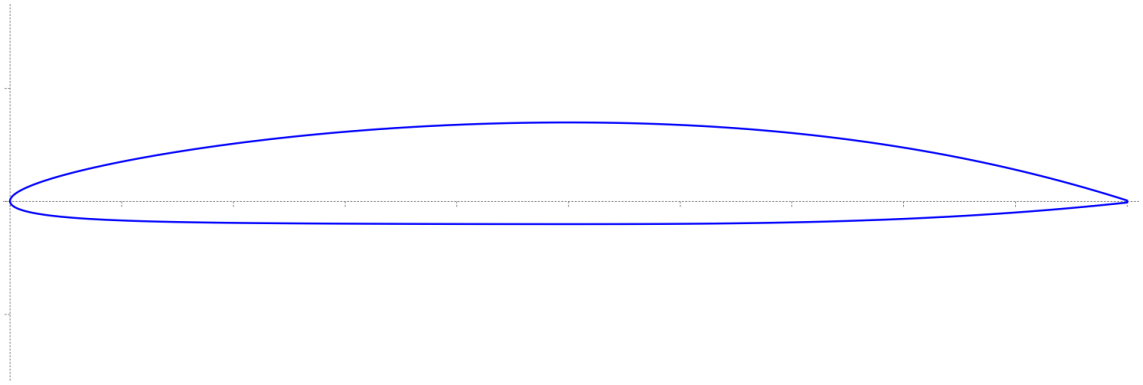

Figure 12: NACA 16-509

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